Harmful Algae 153 (2026) 103068

Contents lists available at ScienceDirect

Harmful Algae

journal homepage: www.elsevier.com/locate/hal

Responses of a harmful algal bloom-causing dinoflagellate *Karenia mikimotoi* to elevated temperature and high N:P ratio

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ARTICLE INFO

Keywords:
Dinoflagellates
Karenia mikimotoi
Harmful algal bloom
Ocean warming
High N:P ratio
Transcriptomics

ABSTRACT

Marine dinoflagellates are increasingly exposed to concurrent ocean warming and eutrophication; however, their responses to multiple concurrent stressors remain inadequately understood. Here, we investigated the individual and combined effects of elevated temperature (26 °C vs. 22 °C) and a high nitrogen-to-phosphorus ratio (180:1) on the harmful algal bloom-causing dinoflagellate *Karenia mikimotoi* during a 30-day exposure experiment. Elevated temperature and high N:P ratio, individually and in combination, significantly increased growth rate by 16–24 % relative to the control, while chlorophyll a (Chl *a*) content decreased by 22–29 %. Cellular particulate organic nitrogen (PON) content declined by 21–22 % under elevated temperature alone and in combination with a high N:P ratio, whereas particulate organic carbon (POC) content remained unchanged across all treatments. Interaction analyses revealed antagonistic effects of elevated temperature and high N:P ratio on growth rate, Chl *a*, and PON contents. Transcriptomic analyses showed that elevated temperature primarily upregulated genes associated with energy production and lipid biosynthesis, whereas a high N:P ratio enhanced the expression of genes involved in nitrogen assimilation and urea cycle-related pathways. Under combined stress, gene expression patterns indicated a transcriptionally inferred shift in energy allocation toward soluble sugar and lipid metabolism, accompanied by downregulation of urea cycle-related genes, suggesting a trade-off between energy conservation and nitrogen utilization. These results highlight the importance of antagonistic interactions between warming and nutrient imbalance in shaping the physiological and molecular responses of dinoflagellates, with implications for predicting harmful algal bloom dynamics under future ocean conditions.

1. Introduction

Dinoflagellates are a major group of phytoplankton widely distributed in coastal and oceanic waters (De Vargas et al., 2015). They contribute significantly to marine primary production and food webs (Taylor et al., 2008; Accoroni et al., 2017; Muhamad et al., 2018). Additionally, they are also the main causative agents of harmful algal blooms (HABs), which pose substantial threats to marine ecosystems, aquaculture, tourism, and coastal infrastructure (Smayda, 2003; Anderson et al., 2012; Wang et al., 2014; Deng et al., 2023). Increasing evidence suggests that elevated temperature (Gobler et al., 2017; Ou et al., 2017) and eutrophication driven by excessive phosphorus and nitrogen (Edwards et al., 2006; Xiao et al., 2018) are two major stressors in coastal ecosystems driven by climate change and anthropogenic activities. These factors have influenced the geographic distribution, cell abundance, and compositional ratios of dinoflagellates in the ocean, thereby affecting the intensity, scale, and frequency of dinoflagellate-causing HABs (Anderson and Burkholder, 2002; Edwards et al., 2006; Gobler et al., 2017; Gobler, 2020). Therefore, elucidating how marine dinoflagellates respond to elevated temperature and eutrophication has become a central focus in HAB research.

Elevated temperature and eutrophication significantly influence the growth and physiology of dinoflagellates. Elevated temperatures enhance cell growth rate (Lim et al., 2006; Brandenburg et al., 2017),

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https://doi.org/10.1016/j.hal.2026.103068
Received 25 November 2025; Received in revised form 14 January 2026; Accepted 28 January 2026
Available online 29 January 2026
1568-9883/

augment toxin production (Lim et al., 2006), and reduce contents of the particulate organic carbon (POC) and particulate organic nitrogen (PON) (Zhang et al., 2024). Additionally, certain dinoflagellates cope with thermal stress by regulating oxidative stress, homeostatic regulation, fatty acid metabolism, and amino acid transport (Deng et al., 2015; Kang et al., 2021). A high nitrogen-to-phosphorus (N:P) ratio leads to alterations in the substance composition related to growth in *Alexandrium tamarense* (Leong and Taguchi, 2004), notably increasing nitrogen-rich biochemical components such as proteins and paralytic shellfish poisoning (PSP) toxins (Murata et al., 2006, 2012). It also increases the cell size of *A. fundyense*, leading to increased arginine production for PSP synthesis, and affects the cell growth rate, nitrogen uptake and assimilation, as well as the proportion of nitrogenous compounds in *Prorocentrum obtusidens* and *P. minimum* (Li et al., 2011). Notably, the alterations induced by multiple stressors, including elevated temperatures and eutrophication, often co-occur (Bopp et al., 2013) and may have complex, cumulative effects on marine dinoflagellates (Xiao et al., 2018). Several studies have investigated the combined effects of elevated temperatures and eutrophication on phytoplankton, revealing their synergistic interactions. For instance, elevated temperatures affect the composition and structure of algal cells, altering the N:P ratio (Shuter, 1979; Raven and Geider, 1988; Thrane et al., 2017; Armin and Inomura, 2021), whereas a high N:P ratio modifies the optimal temperature range for cells (Gerard, 1997; Wiedenmann et al., 2013; Thomas et al., 2017). Recent research indicates that elevated temperature and high N:P ratio exert synergistic positive effects on the cell growth, Fv/Fm, and C:N ratio of *P. obtusidens*, while exhibiting a synergistic negative effect on PON content (Zhang et al., 2024). However, our understanding of the combined effects of these stressors on dinoflagellates remains limited, constraining our ability to accurately predict their responses to multiple environmental stressors, such as elevated temperatures and eutrophication. Consequently, understanding how dinoflagellates respond to a combination of elevated temperature and eutrophication could improve our assessment of their future behavior, providing a more realistic projection of the ecological consequences of climate change on marine dinoflagellates.

Karenia mikimotoi is a well-documented ichthyotoxic dinoflagellate species responsible for HABs globally, leading to mass mortality of fish, shellfish, and other invertebrates in coastal waters, including those in Asia (Oda, 1935; Park et al., 2013; Li et al., 2017), Europe (Pingree et al., 1975; Crespo et al., 2007; Davidson et al., 2009), Oceania (Chang, 1996), and Africa (Carreto et al., 2001). Previous studies have shown that elevated temperature and high N:P ratio influence the cell growth, physiology, and metabolism of *K. mikimotoi*. Elevated temperatures have been shown to enhance cell growth rate (Munir et al., 2015), increase nitrate assimilation capacity (Dai et al., 2014), elevate photosynthesis rate, and up-regulate the expression of the ribulose-1,5-bisphosphate carboxylase/oxygenase gene compared with other competitors, such as *P. obtusidens* (Shen et al., 2016). A high N:P ratio affects the cell growth rate (Pan and Wu, 2017), chlorophyll *a* (Chl *a*) content (Sun et al., 2004), carotene content, and superoxide dismutase activity (Guan and Li, 2017) of *K. mikimotoi*. In this study, we investigated the physiological, biochemical, and molecular responses of *K. mikimotoi* to elevated temperature, high N:P ratio, and their combination to gain a more comprehensive understanding of dinoflagellates under these conditions. Our findings reveal the antagonistic effects of combined elevated temperature and high N:P ratio.

2. Materials and methods

2.1. Organism and culture conditions

The dinoflagellate *K. mikimotoi* was obtained from the Center for Collections of Marine Algae at Xiamen University, China. The cells were maintained in L1 medium (Guillard and Ryther, 1962) at 22 °C under a 14/10 h light/dark cycle with an irradiance of 53 μmol photons m^{-2} s^{-1}. The experimental seawater, with a salinity of 35‰, was collected from the South China Sea and contained 0.1 μmol L^{-1} phosphate and 1 μmol L^{-1} nitrate. This seawater was filtered using a 0.2 μm pore size filtration cartridge (Whatman™), diluted with Milli-Q water to achieve a final salinity of 30‰ to facilitate optimal growth of *K. mikimotoi*, autoclaved at 121 °C for 20 min, and subsequently cooled to room temperature. Prior to the experiments, well-cultivated *K. mikimotoi* cells were transferred to nutrient-free filtered seawater for two days to deplete intracellular nitrogen and phosphorus reserves, as described by Zhang et al. (2024). Notably, the *K. mikimotoi* culture used in this study was not axenic.

2.2. Experimental design

A two-factor design was employed to investigate the interactive effects of elevated temperature and a high N:P ratio on *K. mikimotoi*. Each factor had two levels: current conditions, represented by a seawater temperature of 22 °C and N:P ratio of 40:1, and projected future conditions, with a seawater temperature of 26 °C and N:P ratio of 180:1. The current temperature (22 °C) corresponds to the maintenance temperature of the culture. The future scenarios are based on the global change projections for 2100 (Xiao et al., 2018; IPCC, 2023). The experimental setup comprised four treatments: “C” (control: seawater temperature of 22 °C and N:P ratio of 40:1), “T” (elevated temperature: 26 °C, 40:1), “N” (high N:P ratio: 22 °C, 180:1), and “TN” (26 °C, 180:1), with three biological replicates for each treatment.

To achieve the low and high N:P ratios, the nitrogen concentration was adjusted while maintaining a constant phosphorus concentration of 1 μmol L^{-1}. Temperature treatments corresponding to current and future temperature scenarios were controlled using a dual-control temperature incubator. The cells were grown in a semi-continuous culture manner, with dilution every three days by transferring to fresh seawater (800 mL) at an initial cell density of 1000 cells mL^{-1}. The experiment lasted 30 days (ten growth cycles), during which the cells were sampled every three days to measure the cell growth rate, Chl *a* content, and maximum photosynthetic efficiency (Fv/Fm). Additionally, POC and PON samples were collected every six days. At the end of the experiment (day 30), cells of each treatment were collected for transcriptome analysis, adenosine triphosphate (ATP) content, amino acid (AA) content, and activity analyses of key enzymes related to carbon assimilation (carbonic anhydrase (CA) and phosphoenolpyruvate carboxykinase (PEPCK)) and nitrogen assimilation (nitrate reductase (NR) and glutamine synthetase (GS)).

2.3. Physiological parameter analysis

Cell density was determined using an optical microscope. The growth rate was calculated using the formula $\mu = (\ln N_1 - \ln N_0) / (t_1 - t_0)$, where N_1 and N_0 represent cell densities at the beginning (t_0) and the end (t_1) of each semi-continuous run. The Fv/Fm value was measured before each dilution using a Phyto-PAM Phytoplankton Analyzer (Phyto-PAM, Heinz Walz GmbH). For Chl *a* content, cells were filtered onto a GF/F filter (25 mm, Whatman) and extracted overnight with 90 % acetone at –20 °C in dark. The extract was subsequently centrifuged, and the Chl *a* concentration in the supernatant was quantified using a Turner Trilogy fluorometer (Turner Designs, USA).

To determine the POC and PON contents, cells were filtered onto a pre-combusted (450 °C, 6 h) GF/F filter (25 mm, Whatman). The filters were exposed to HCl fumes for 12 h and then dried overnight at 60 °C. POC and PON contents were quantified using an Elemental Analyzer (2400; Series II CHNS/O; PerkinElmer, Waltham, MA, USA).

Protein content was determined using a protein content assay kit (Biuret Method) (Solarbio, China; Cat. No. BC3185), following the manufacturer’s instructions. For the measurement of ATP and AA contents, as well as enzyme activities, including nitrate reductase (NR), glutamine synthetase (GS), carbonic anhydrase (CA), and phosphoenolpyruvate carboxykinase (PEPCK), cells were filtered onto 3 μm polycarbonate (PC) membranes and analyzed using specific assay kits (Solarbio, China; Cat. No. BC0300, BC1575, BC0085, BC0915, BC5440, and BC3315). All enzyme activities and metabolite contents were normalized to the total protein content of the sample.

2.4. Transcriptome analysis

The cells were filtered onto a 3 μm PC membrane. Total RNA was extracted from *K. mikimotoi* cells using TRIzol® Reagent (TaKaRa Biotechnology, Shiga, Japan) according to the manufacturer's instructions. RNA integrity was evaluated using an RNA Nano 6000 Assay Kit on a bioanalyzer (2100 system; Agilent Technologies, Santa Clara, CA, USA). Before RNA sequencing, messenger (m)RNA was isolated from total RNA using poly T oligo-attached magnetic beads. Paired-end complementary (c)DNA libraries were constructed using random oligonucleotides as primers. RNA sequencing was performed using the NovaSeq™ 6000 platform (Illumina, San Diego, CA, USA).

High-quality clean reads were obtained by eliminating those containing adapters, reads with N bases, and low-quality reads from the raw data. The Q20 of the clean reads exceeded 95 % for all the samples. High-quality clean reads were assembled into unigenes using a *de novo* assembler (Trinity software). Unigenes were annotated against the National Center for Biotechnology Information Non-Redundant (NCBI-NR), National Center for Biotechnology Information Nucleotide (NCBI-NT), Swiss-Prot, Clusters of Orthologous Genes (COG), Gene Ontology (GO), and Kyoto Encyclopedia of Genes and Genomes (KEGG) databases using BLAST (v2.2.26 + x64-linux). The gene expression of the assembled unigenes was normalized using fragments per kilobase per million fragments (FPKM). To analyze the differences in gene expression between the three treatments (T, N, and TN) and the control (C), $|\log_2$ (fold change)$| > 1$ and P-value < 0.05 were used as thresholds. Analyses of the enrichment of function and signaling pathways of differentially expressed genes (DEGs) were performed using the GO (https://geneontology.org/) and KEGG (www.genome.jp/) databases. A Q-value < 0.05 was set to identify significantly enriched pathways.

2.5. Validation of differentially expressed genes (DEGs)

To validate the transcriptome results, real-time reverse transcription quantitative polymerase chain reaction (RT-qPCR) was employed to quantify the expression of 20 DEGs associated with important metabolic pathways and cellular functions, including photosynthesis, carbohydrate metabolism, lipid metabolism, and nitrogen metabolism (Table S1). Glyceraldehyde-3-phosphate dehydrogenase (*GAPDH*) was used as the internal standard (Shi et al., 2020). cDNA was synthesized from mRNA using the PrimeScript RT reagent kit (Takara, Japan), and RT-qPCR was performed using the TB Green Premix Ex Taq (Takara, Japan). The primer sequences are listed in Table S1. Data reliability was ensured by analyzing each gene in three biological replicates and two technical replicates. Relative gene expression was calculated using the $2^{-\Delta\Delta t}$ method (Schmittgen and Livak, 2008; Zhang et al., 2019a). The correlation between the RNA sequencing and RT-qPCR results was assessed using regression analysis.

2.6. Statistical analyses

All data are presented as the mean ± standard deviation. Statistical analyses were conducted using IBM SPSS Statistics software (version 25). A one-way analysis of variance (ANOVA) was employed to evaluate the differences in physiological parameters among the groups (C, N, T, and TN), with a significance threshold set at $P < 0.05$. This analysis was used solely to assess the overall differences among the treatment groups rather than to test statistical interactions.

Folt's method was used to examine the interaction between the two stressors (Folt et al., 1999). The observed effect (*OE*) was determined as the percentage change in the treatment group compared to that in the control group. The additive effect (*AE*) of multiple stressors is defined as $AE=E_1+E_2+\cdots+E_n$. The multiplicative effect (*ME*) is defined as $ME=(1+E_1)\times(1+E_2)\times\cdots\times(1+E_n)$-1, where En represents the individual effect of stressor n on the measured physiological metric, calculated as the percentage change in the physiological metric at the predicted condition for the year 2100 relative to the present condition.

The interactions among stressors were categorized as follows:

(1) Additive effect: the *OE* of two or more environmental drivers is equivalent to the calculated *AE*.
(2) Synergistic interactive effect (or synergism): the *OE* of two or more environmental stressors exceeds the calculated *ME*, which can be either positive or negative.
(3) Antagonistic interactive effect (or antagonism): the *OE* of two or more environmental stressors is less than the calculated *ME*, which can be either positive or negative (Folt et al., 1999; Boyd and Hutchins, 2012; Boyd et al., 2018).

3. Results

3.1. Physiological and biochemical responses of K. mikimotoi

The average growth rates of *K. mikimotoi* were 1.16-, 1.17-, and 1.24-fold higher in the T, N, and TN groups, respectively, than that in the control group (0.38 ± 0.07 d^{-1}) ($P < 0.05$, Fig. 1A). Chl *a* concentration decreased by 1.22-, 1.25-, and 1.29-fold in the T, N, and TN groups, respectively, relative to the control group (12.27 ± 2.55 pg cell^{-1}) ($P < 0.05$, Fig. 1C). The cellular PON content was reduced by 1.22- and 1.21-fold in the T and TN groups, respectively, compared to that in the control group (158.72 ± 19.47 pg cell^{-1}) ($P < 0.05$, Fig. 1E). No significant variations were observed in Fv/Fm, cellular POC content, or the molar ratio of POC:PON (or organic C:N ratio) across all treatments (Fig. 1B, D, and F).

The activity of NR, an enzyme crucial for nitrate assimilation, did not differ significantly among the treatments (Fig. 2A). However, the activity of GS, another enzyme involved in nitrate assimilation, was 1.24-fold lower in the N group than in the control group (0.22 ± 0.02 U mg^{-1} prot) ($P < 0.05$, Fig. 2B). The activities of CA and PEPCK, both enzymes essential for carbon fixation, showed similar trends; they were 3.07- and 1.60-fold higher in the N group and 2.96- and 1.63-fold higher in the TN group, respectively, compared to the control group (0.36 ± 0.11 U mg^{-1} prot and 169.78 ± 29.66 U mg^{-1} prot) ($P < 0.05$, Fig. 2C-D); however, they did not significantly vary in the T group. The highest ATP content was observed in the N group, which was 2.62-fold higher than that in the control group (0.06 ± 0.02 μmol mg^{-1} prot) ($P < 0.05$, Fig. 2F).

3.2. Interactive effects of elevated temperature and high N:P ratio on the physiology of K. mikimotoi

The interactive effects of T and N on the physiology of *K. mikimotoi* were assessed by comparing their observed effects (*OE*) with their predictive effects (*AE* and *ME*) (Table 1). For the growth rate and contents of Chl *a* and PON, TN had a less pronounced *OE* than *ME*. This indicates that the interaction of T and N had an antagonistic effect on the physiology of *K. mikimotoi*, with the growth rate as the antagonistic positive effect, whereas the contents of Chl *a* and PON exhibited antagonistic negative effects.

3.3. Transcriptomic response of K. mikimotoi

Sequencing of the qualified cDNA libraries was performed using the NovaSeq 6000 platform. After the raw data were filtered and checked, the Q20 and GC values for all samples were 98 % and 53 %, respectively. After *de novo* assembly using Trinity software (http://trinityrnaseq.github.io), 136,524 unigenes were identified, with an average length of

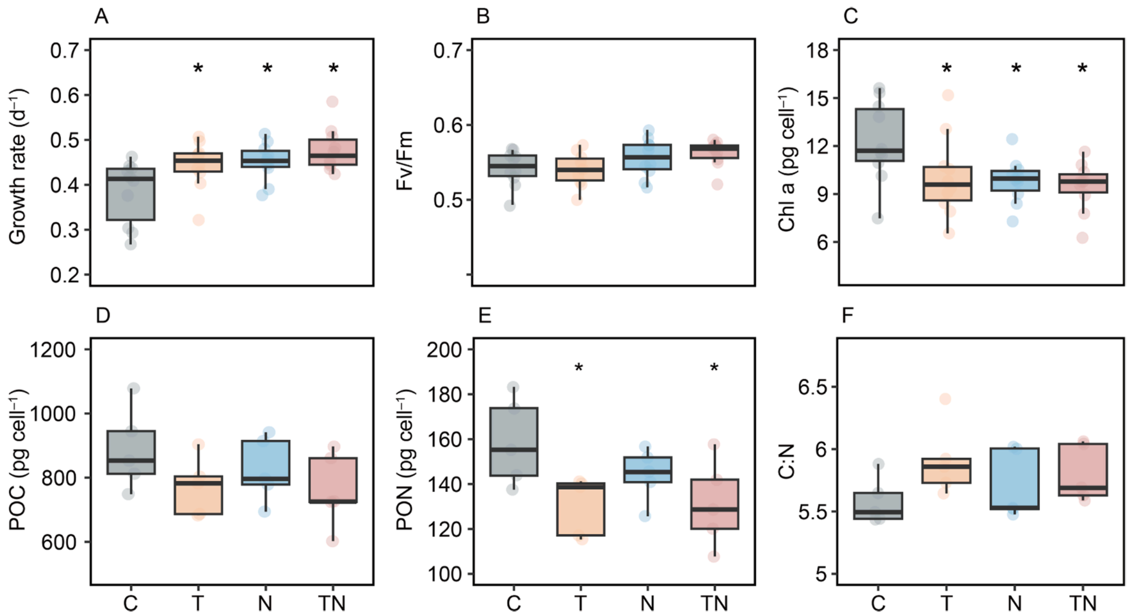

Fig. 1. Effects of elevated temperature and high N:P ratio on physiological traits of *Karenia mikimotoi*. (A) Growth rate, (B) Fv/Fm, (C) chlorophyll *a* (Chl *a*) content, (D) POC content, (E) PON content, and (F) C:N ratio. "*" indicates significant differences from the control.

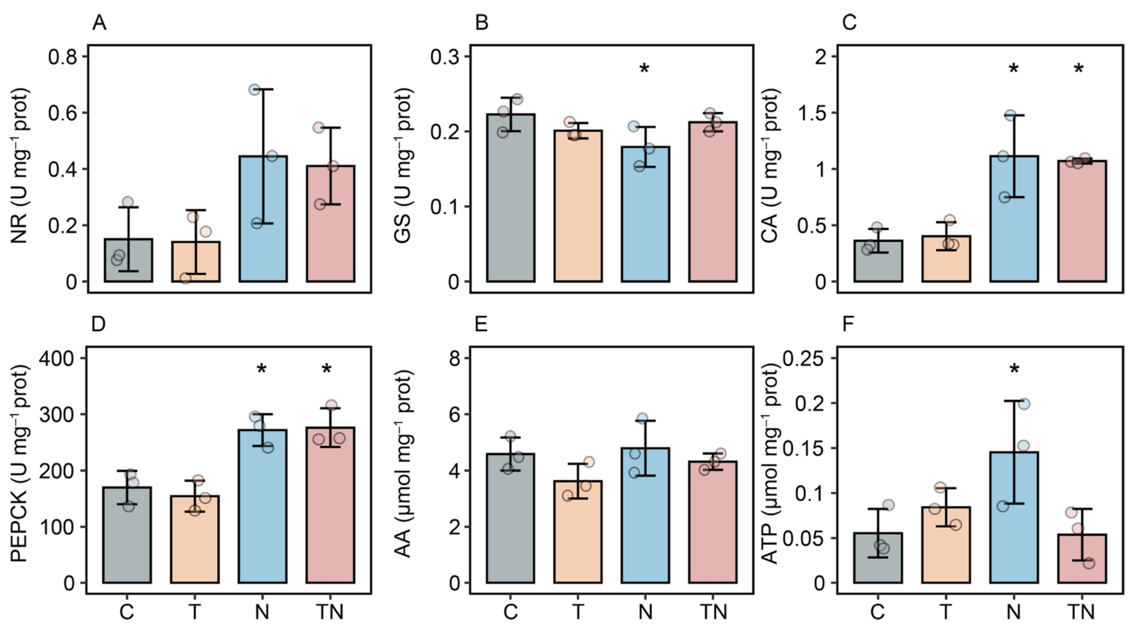

Fig. 2. Effects of elevated temperature and high N:P ratio on biochemical traits of *Karenia mikimotoi*. (A) nitrate reductase (NR), (B) glutamine synthetase (GS), (C) carbonic anhydrase (CA), (D) phosphoenolpyruvate carboxykinase (PEPCK), (E) amino acid (AA), and (F) adenosine triphosphate (ATP). "*" indicates significant differences from the control.

Table 1
Interactive effects of elevated temperature and high N:P ratio on physiological parameters of *K. mikimotoi*.

Physiological traits	Observed effect			Interactive effect		Interactive effects
	OE_T	OE_N	OE_{TN}	AE_{TN}	ME_{TN}	
Growth rate	15.6 %	17.5 %	24.4 %	33.1 %	35.8 %	Antagonistic effect
Chl *a* content	−17.9 %	−19.8 %	−22.8 %	−37.7 %	−34.2 %	Antagonistic effect
POC content	n.s.	n.s.	−14.1 %	n.s.	n.s.	—
PON content	−17.8 %	n.s.	−17.3 %	−17.8 %	−17.8 %	Antagonistic effect

Note: n.s. indicates no significant effects (according to one-way ANOVA analysis).

1194 bp and N50 of 1856 bp (Table S2). Gene function annotation was performed for all unigenes by querying seven databases, resulting in a total of 98,147 unigenes (71.88 %) being annotated in at least one database, specifically NR (77,658, 56.88 %), NT (2669, 1.95 %), KO (21,953, 16.07 %), SwissProt (42,889, 31.41 %), PFAM (78,480, 57.48 %), GO (78,473, 57.47 %), and KOG (17,963, 13.15 %) (Table S3).

Principal component analyses (PCA) revealed a distinct separation in the transcriptome of *K. mikimotoi* between the elevated temperature (T and TN) and ambient temperature (C and N) groups, suggesting that elevated temperature significantly influences gene expression in *K. mikimotoi* (Fig. 3A). The volcano plot of DEGs indicated that 2027, 1205, and 2938 DEGs were identified in the T vs. C, N vs. C, and TN vs. C comparisons, respectively ($|\log_2$ (fold change)$| > 1$ and P-value < 0.05) (Fig. 3B). The Venn diagram (Fig. 3C) shows that the T vs. C and TN vs. C comparisons shared 251 DEGs, whereas the N vs. C and TN vs. C comparisons shared only 75 DEGs. Furthermore, hierarchical clustering analysis revealed that DEGs were distributed in two distinct comparison groups: elevated temperature (T vs. C and TN vs. C) and high N:P ratio (N vs. C) (Fig. 3D).

The KEGG pathway analysis (Fig. 3E) indicated that in the T vs. C comparison, upregulated DEGs were associated with "carbon metabolism", "cell cycle", and "photosynthesis-antenna proteins", whereas downregulated DEGs were linked to "ribosome" (Q-value < 0.05). In the N vs. C comparison, upregulated DEGs were enriched in "photosynthesis-antenna proteins" and "carbon metabolism", while downregulated DEGs were associated with "photosynthesis" (Q-value < 0.05). In the TN vs. C comparison, upregulated DEGs were related to "biosynthesis of amino acids" and "carbon metabolism", and downregulated DEGs to "ribosome". Functional categorization revealed that the DEGs were primarily involved in stress response (Table S4), photosynthesis (Table S5; Fig. 4), carbohydrate metabolism (Table S6; Fig. 4), lipid metabolism (Table S7; Fig. 4), nitrogen assimilation (Table S8; Fig. 4) and protein metabolism (Table S9; Fig. 4).

Twenty DEGs were selected for RT-qPCR validation, demonstrating a strong correlation with the RNA- seq results ($R^2 = 0.76$; $P < 0.05$), thereby confirming the reliability of the transcriptomic data (Fig. 3F).

4. Discussion

4.1. Physiological response of K. mikimotoi to elevated temperature and high N:P ratio

Most algae exhibit enhanced growth at higher temperatures. For example, *K. mikimotoi* presents optimal growth between 19 and 28 °C in the East China Sea (Zhang et al., 2022), with significantly higher growth at 24 °C than at 16 and 20 °C (Shen et al., 2016). Nutrient ratios also influence growth; for instance, the dinoflagellate *A. tamarense* shows increased growth under higher N:P ratios when phosphorus is limited

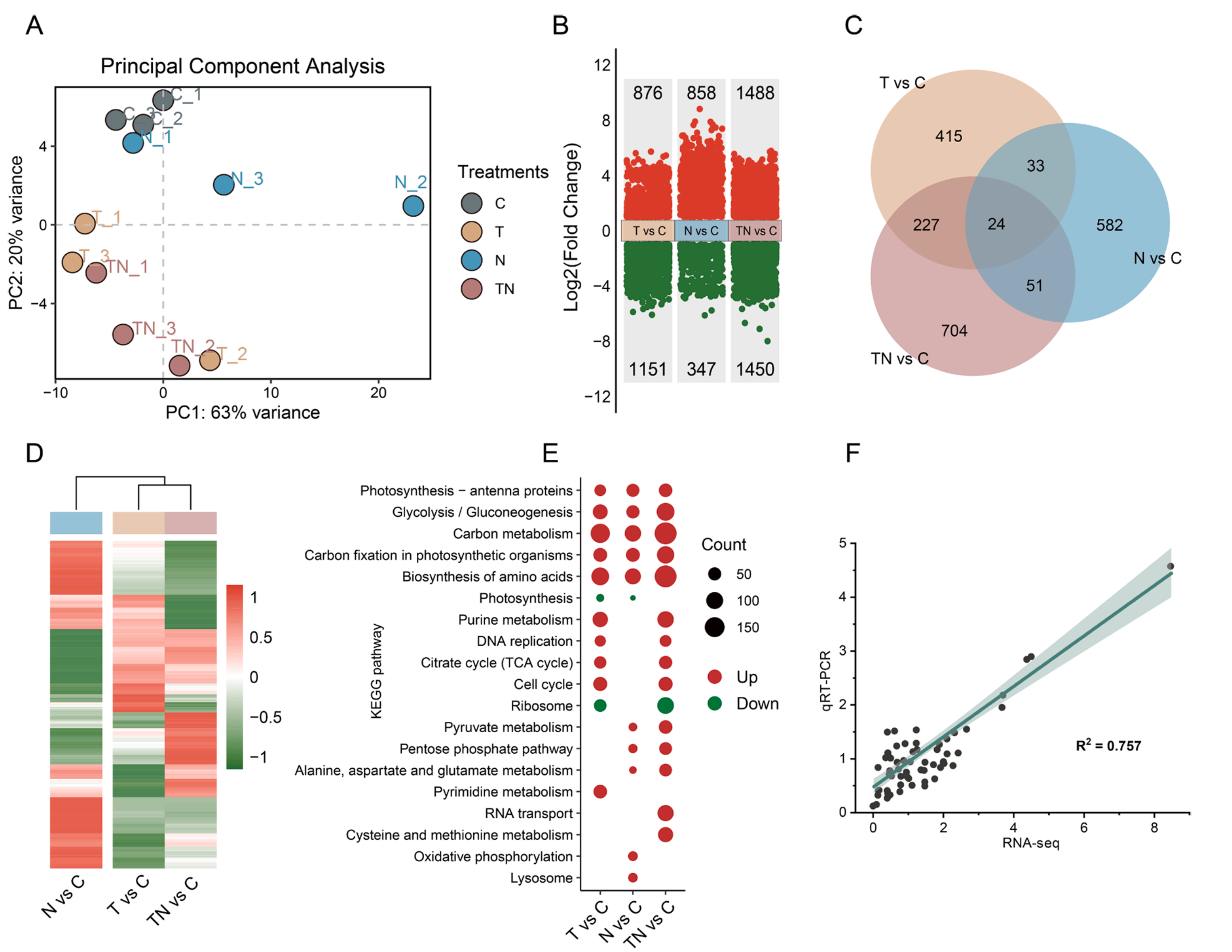

Fig. 3. Transcriptome profiles of differentially expressed genes (DEGs). (A) PCA shows a separation among four experimental treatments. PC1 (63.00 %) and PC2 (20.00 %) were the most and second most abundant underlying variations among samples, respectively. (B) Volcano plot of DEGs ($|\log_2$ (fold change)$| > 1$ and P-value < 0.05) from three comparison groups (T vs C, N vs C, TN vs C). (C) Venn plot of DEGs. (D) Hierarchical clustering analysis was performed for $\log_2$ (fold change) of all DEGs. (E) The KEGG-based enrichment pathways for DEGs in different pair comparisons. and (F) Correlation between RNA-Seq and RT-qPCR data. 20 DEGs were identified using RT-qPCR.

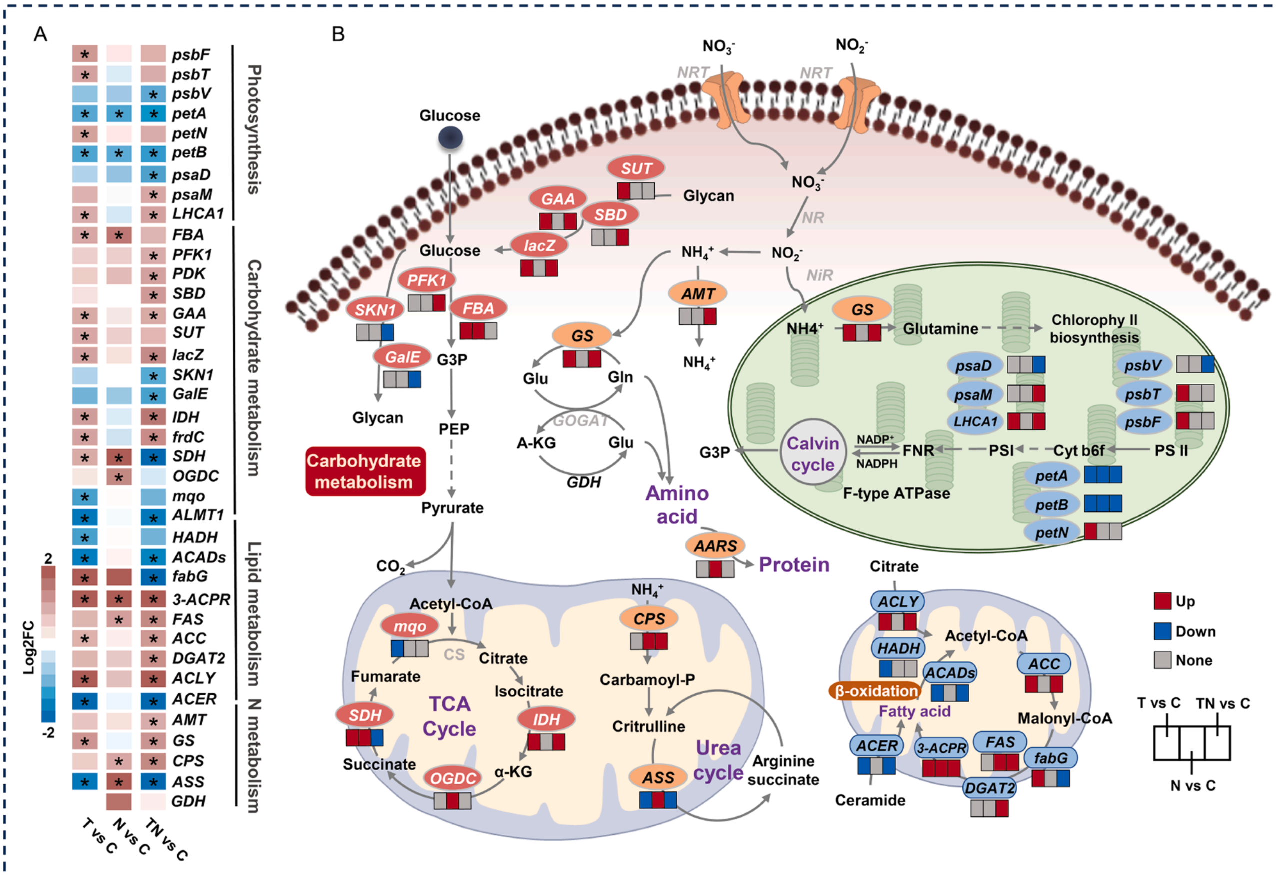

Fig. 4. (A) Key genes and (B) schematic representations of the mechanisms underlying the response of *Karenia mikimotoi* cells to elevated temperature and high N:P.

(Murata et al., 2012). In this study, the growth rate of *K. mikimotoi* significantly increased under elevated temperature and high N:P ratio, particularly when both factors were combined, suggesting potential advantages under future ocean conditions characterized by warming and high N:P ratio. Despite the reduction in Chl *a* content, which was likely due to accelerated cell division, Fv/Fm remained stable, indicating a sustained photosynthetic efficiency. These findings suggest that *K. mikimotoi* may thrive in warmer and/or high N:P ratio marine environments, potentially leading to more frequent blooms by 2100. This contrasts with our previous findings on the dinoflagellate *P. obtusidens* (Zhang et al., 2024), where a high N:P ratio had no significant effect on growth, suggesting that *P. obtusidens* and *K. mikimotoi* respond differently to high N:P ratios, which might be caused by their different cell morphologies. *P. obtusidens* is armored, while *K. mikimotoi* is unarmored. This difference may reflect the stronger phagotrophic capacity of unarmored dinoflagellates (Zhang et al., 2013), potentially conferring a competitive advantage in future oceans with high N:P ratios.

Temperature and nutrients are key regulators of phytoplankton stoichiometry at both species and community levels (Boyd et al., 2015; Feng et al., 2020; Zhang et al., 2024). In this study, elevated temperature did not significantly affect the POC content in *K. mikimotoi*, but it significantly decreased the PON content, likely due to accelerated cell division. Cellular carbon is essential for the synthesis of light-harvesting proteins, biosynthesis, and energy storage (Geider et al., 1996). Elevated temperatures may enhance the synthesis of carbon-rich compounds for stress protection, as observed in *Effrenium voratum* (Symbiodiniaceae) (Yang et al., 2022) and *P. obtusidens* (Shen et al., 2016). This could account for the stable POC content despite the increased growth rates. Conversely, the reduction in PON content at elevated temperatures may indicate decreased nitrogen allocation to pigments and proteins, which is consistent with the findings that elevated temperatures increase the C:N ratio in *P. shikokuense* (Zhang et al., 2022). Under conditions of a high N:P ratio, *K. mikimotoi* exhibited faster growth without significant changes in POC or PON content, suggesting enhanced synthesis of nitrogen-containing organic compounds, which helps cells cope with high N:P ratio stress. As nitrogen assimilation relies on the carbon skeleton provided by photosynthesis (Foyer et al., 2001), the POC content was not inhibited in the present study. Similarly, under high nitrogen concentrations, nitrogen-rich organic compounds, such as proteins and PSP, significantly increased in *A. tamarense* (Murata et al., 2012). Under the combined effect of elevated temperature and a high N:P ratio, the POC content remained unchanged, whereas the PON content decreased significantly. Notably, the combined effect of elevated temperature and high N:P ratio on PON content revealed an antagonistic effect, indicating that elevated temperature inhibits the accumulation of nitrogen-containing organic compounds in *K. mikimotoi* at high N:P ratios. Armin and Inomura (2021) proposed that increased enzymatic activity and catalytic efficiency at higher temperatures reduce the demand for nitrogen-rich proteins, leading to lower nitrogen reserves and increased carbon storage. Overall, the observed stoichiometric shifts suggest that *K. mikimotoi* regulates its elemental composition to cope with environmental stress, a strategy that supports its survival and growth and may influence marine primary productivity.

4.2. Transcriptomic response of K. mikimotoi to elevated temperature and high N:P ratio

4.2.1. Stress response

Phytoplankton typically exhibit a stress response following exposure to external environmental stressors. This response is often associated with an imbalance between oxidative and antioxidant systems, leading to excessive intracellular production of highly reactive molecules, such as reactive oxygen species (ROS), which can adversely affect cellular function (Zhang et al., 2016). To counteract ROS-induced damage, cells activate antioxidant enzymes, including serine/threonine protein kinase (AKT), mitogen-activated protein kinase 6 (MPK6), glutathione peroxidase (GSH-Px), glutathione S-transferase (GST), peroxidase (POD), and catalase (CAT) (Mittler et al., 2022). In this study, genes associated with antioxidant defense were upregulated under elevated temperature (*AKT, MPK6, POD*, and *CAT*), high N:P ratio (*GST* and *CAT*), and their combination (*AKT, MPK6, POD, CAT*, and *GSH-Px*) (Table S4), indicating significant oxidative stress, particularly at elevated temperatures. Similar patterns have been observed in *Chlamydomonas reinhardtii* (Chlorophyta), which enhances antioxidant gene expression under long-term heat stress (Hemme et al., 2014). Calcium signaling, a critical stress response pathway (Lecourieux et al., 2006), was activated. Genes encoding calcium-binding proteins and calcium channel proteins were upregulated under elevated temperature alone and in combination with a high N:P ratio, indicating a calcium-mediated thermal stress response. Notably, the calcium-dependent protein kinase gene, a primary receptor for calcium ion signaling, was downregulated under the combined effect of elevated temperature and high N:P ratio, suggesting that a high N:P ratio may have mitigated calcium-mediated signaling responses, whereas oxidative stress-related responses remained increased. Overall, *K. mikimotoi* experienced excessive oxidative stress under all treatments, with temperature being the primary driver, and a high N:P ratio may exert partial mitigation via specific signaling pathways. Oxidative stress has been linked to dinoflagellate cytotoxicity through redox-related processes, including hemolytic activity and ROS dynamics (Ou et al., 2017), and integrated physiological and transcriptomic analyses further indicate a direct association between temperature-induced oxidative stress and enhanced cytotoxicity (Shi et al., 2025). Accordingly, the upregulation of antioxidant related genes under combined stress suggests shifts in cellular redox balance that may indirectly modulate toxin-related processes.

4.2.2. Photosynthesis

Photosynthesis allows phytoplankton to synthesize organic matter using light energy harvested mainly by photosystems I and II, together with the cytochrome b6f complex and ATP synthase (Whitmarsh and Govindjee, 1999). The cytochrome b6-f complex facilitates electron flow between photosystems I and II, aiding in the conversion of light energy into a transmembrane proton gradient for ATP production (Malone et al., 2019). In this study, elevated temperature upregulated photosystem II components (e.g., photosystem II cytochrome b559 subunit beta and photosystem II PsbT protein), indicating enhanced light responses in the photosystem II reaction center. However, under the combined effect of elevated temperature and a high N:P ratio, genes related to the cytochrome b6-f complex (e.g., apocytochrome f and cytochrome b6) were downregulated, suggesting a diminished electron transport efficiency. This finding aligns with the hypothesis that plants adjust nitrogen allocation to sustain photosynthetic efficiency under stress. For example, plants allocate more nitrogen to the cytosol under low-irradiance conditions, which increases the number of cytochromes in the b6-f complex and decreases the electron transport capacity per unit of chlorophyll (Evans, 1989). Regarding photosystem I, the down-regulation of photosystem I subunit II (*psaD*), which provides proteins for electron transfer reaction, and the up-regulation of photosystem I subunit XII (*psaM*), which maintains the stability of the light-trapping antenna protein dimer, along with increased expression of light-harvesting chlorophyll a/b binding protein under the combined effect of elevated temperature and high N:P ratio, suggest reduced electron transfer but enhanced light-harvesting capacity (Yan et al., 2021). These transcriptional changes indicate that *K. mikimotoi* may cope with elevated temperatures alone and in combination with a high N:P ratio by reallocating resources to optimize light capture rather than electron transport, a trade-off strategy that likely maintains energy acquisition and supports growth while reducing nitrogen demand under suboptimal conditions.

4.2.3. Carbohydrate metabolism

Elevated temperature and high N:P ratios affect the energy and carbohydrate metabolism of phytoplankton. Previous studies have shown that *K. mikimotoi* up-regulates glycolysis and the tricarboxylic acid (TCA) cycle at higher nitrogen concentrations (Shi et al., 2021), whereas *P. obtusidens* enhances energy production through glycolysis, the TCA cycle, and nitrogen and carbon assimilation at elevated temperature (Zhang et al., 2024). In the present study, genes associated with glycolysis, such as fructose 1,6-bisphosphate aldolase (under elevated temperature and a high N:P ratio) and 6-phosphofructokinase 1 (under the combined effect of elevated temperature and a high N:P ratio), were upregulated, indicating increased energy demand. TCA cycle genes, including succinate dehydrogenase (*SDH*) and 2-oxoglutarate dehydrogenase, were also upregulated under a high N:P ratio, consistent with the higher ATP content (Fig. 2F), suggesting enhanced mitochondrial energy production to support accelerated growth and metabolic activity. Notably, temperature and nutrient conditions affected specific TCA cycle branches. Under elevated temperature, isocitrate dehydrogenase (*IDH*) and *SDH* were upregulated, whereas malate quinone oxidoreductase was downregulated. When elevated temperature was combined with a high N:P ratio, the expression of *IDH* and fumarate reductase subunit C was up-regulated, whereas *SDH* was down-regulated. These patterns suggest that *K. mikimotoi* may enhance energy generation via glycolysis and the oxidative decarboxylation steps in the first half of the TCA cycle (Martínez-Reyes and Chandel, 2020), while down-regulating the second half through feedback to prevent energy overloading. This shift may lead to the accumulation of intermediates, such as citric acid, which could be redirected toward fatty acid synthesis (Yang et al., 2022), reflecting a flexible energy strategy that enables dinoflagellates to balance cellular growth, energy production, and storage under multiple stress conditions.

Elevated temperatures can disrupt the osmotic balance of plant cells, with soluble sugars serving as solutes to regulate intracellular osmotic pressure, thereby enabling cells to store soluble sugars as a defense mechanism against environmental stress (Ruan et al., 2010). In this study, elevated temperature upregulated genes associated with polysaccharide hydrolysis, including alpha-glucosidase (*GAA*), sucrose transporter (*SUT*), and beta-galactosidase (*lacZ*). The combined effect of elevated temperature and a high N:P ratio also upregulated polysaccharide hydrolysis genes, such as starch binding domain, *GAA, SUT*, and lacZ, but downregulated genes involved in polysaccharide synthesis, such as β-glucan synthesis-associated protein and UDP-glucose 4-epimerase. These transcriptional changes in polysaccharide hydrolysis and synthesis related genes suggest a potential redistribution of carbon from structural polysaccharides toward soluble sugars, which may help *K. mikimotoi* regulate its osmotic balance and mitigate heat stress (Ruan et al., 2010). Overall, both elevated temperature and a high N:P ratio altered carbohydrate metabolism in *K. mikimotoi*, affecting intracellular carbon allocation.

4.2.4. Lipid metabolism

Research on algae predominantly indicates that lipids serve as a more prevalent form of energy storage than carbohydrates. Consequently, algae tend to synthesize more lipids for carbon and energy storage when subjected to environmental stressors (Maltsev and Maltseva, 2021). For example, *E. voratum* shifts energy storage from carbohydrates to lipids during periods of thermal stress (Yang et al., 2022). In this study, genes involved in fatty acid degradation (e.g., 3-hydroxyacyl-CoA dehydrogenase and acyl-CoA dehydrogenases, *ACADs*) were downregulated under elevated temperature alone and in combination with a high N:P ratio. Conversely, genes related to fatty acid synthesis, including acetyl-CoA carboxylase, were upregulated. These results, consistent with the enhanced activity in the first half of the TCA cycle, suggest a potential shift in carbon allocation toward lipid synthesis at the transcriptional level under warming stress to support rapid cell division, which could contribute to the stable intracellular POC levels. In contrast, under similar conditions, *P. obtusidens* upregulates genes involved in fatty acid degradation and reduces POC content (Zhang et al., 2024), indicating divergent metabolic strategies in response to warming between the two species.

Sphingolipid metabolism also demonstrated a responsiveness. Alkaline ceramidase (*ACER*), which hydrolyzes ceramides into sphingosine and fatty acids (Xu et al., 2021), was downregulated under elevated temperature alone and in combination with a high N:P ratio. Given that sphingosine and ceramides influence cell membrane structure and signal transduction (Michaelson et al., 2016), this suggests alterations in membrane composition. Similarly, fatty acid desaturase (*FAD*), which catalyzes unsaturated fatty acid synthesis (Calhoun et al., 2021), was significantly downregulated, consistent with increased membrane saturation to enhance the thermal stability (Murata and Los, 1997; Hixson and Arts, 2016). This shift reflects homeoviscous adaptation, a mechanism that maintains membrane fluidity under stress (Sinensky, 1974), which has been reported in various algae, including *C. reinhardtii* (Hemme et al., 2014), *Symbiodinium* (dinoflagellates) (Kneeland et al., 2013), *Scenedesmus obliquus* (chlorophyta) (Fuschino et al., 2011), and *Odontella aurita* (diatoms) (Pasquet et al., 2014). Overall, elevated temperatures may impair membrane function, prompting *K. mikimotoi* to adjust its lipid composition by downregulating *ACER* and *FAD* as a potential regulatory strategy (Quinn, 1988; Uemura et al., 2006; Kuypers et al., 2018).

4.2.5. Nitrogen metabolism

In this study, nitrate served as the sole nitrogen source for *K. mikimotoi*, necessitating its reduction to ammonia before assimilation into organic nitrogen compounds (Kuypers et al., 2018). Under a high N:P ratio, genes involved in urea metabolism, such as carbamoyl-phosphate synthase (*CPS*) and arginosuccinate synthase (*ASS*), as well as those associated with nitrogen assimilation, including glutamate dehydrogenase (*GDH*), were upregulated, indicating an enhancement of nitrogen assimilation to support growth. Notably, *CPS*, which encodes the first-step enzyme of the urea cycle, was transcriptionally upregulated, indicating enhanced expression of urea cycle-related genes and suggesting a potential modulation of nitrogen recycling pathways at the transcriptional level (Allen et al., 2011). Meanwhile, the upregulation of *GDH* suggests an increased conversion of ammonia to glutamate, a crucial step in nitrogen assimilation (Robinson et al., 1991). Previous studies have underscored the significance of the urea cycle in algal nitrogen metabolism. For example, the ornithine-urea cycle in diatoms not only facilitates the assimilation of carbon into nitrogenous compounds but also modulates the cellular carbon and nitrogen balance (Allen et al., 2011). Other algae, such as *Zooxanthella*, can rapidly store nitrogen as uric acid crystals in response to sudden nitrogen increases (Kopp et al., 2013). However, under elevated temperature alone and in combination with a high N:P ratio, the expression of genes encoding *ASS* was downregulated, while *GS* was upregulated, suggesting a weakening of the urea cycle and a potential shift towards the GS/GOGAT pathway for ammonia assimilation. This transition may be related to the energetic advantages of the GS/GOGAT cycle, which, as discussed by Lancien et al. (2000), represents a highly efficient route for ammonium assimilation in plants. Concurrently, under the combined effect of elevated temperature and high N:P ratio, the upregulation of ammonium transport and *CPS* genes suggests increased production of carbamoyl phosphate, which may be redirected from the urea cycle to pyrimidine nucleotide synthesis (Shi et al., 2018), thereby facilitating efficient nitrogen utilization by *K. mikimotoi* to support growth and reproduction.

Aminoacyl-tRNA synthetases (*AARS*), which are essential for protein biosynthesis by charging tRNAs with amino acids (Rajendran et al., 2018), exhibited differential expression under varying conditions. Specifically, only genes encoding phenylalanyl-tRNA synthetase were upregulated under elevated temperature alone and in combination with a high N:P ratio, while genes for threonyl-, tryptophanyl-, histidyl-, and arginyl-tRNA synthetase were upregulated under a high N:P ratio, indicating an increased demand for specific amino acids and enhanced protein synthesis. Similarly, elevated temperatures increased the essential amino acid content in the diatom *Cylindrotheca fusiformis* (Bermúdez et al., 2015), and *AARS* gene expression in *A. catenella* correlated positively with nitrate levels (Saldivia et al., 2023). Collectively, these results suggest that elevated temperature, high N:P ratio, and their combination alter the nitrogen metabolism in *K. mikimotoi*, enabling flexible nitrogen allocation that supports growth, stress responses, and reproduction in changing marine environments.

4.3. Antagonistic interaction between elevated temperature and high N:P ratio on K. mikimotoi

Marine phytoplankton often exhibit synergistic or antagonistic responses to multiple environmental drivers due to molecular regulation and energy trade-offs (Schlüter et al., 2014). Here, we observed an antagonistic interaction between elevated temperature and a high N:P ratio in *K. mikimotoi*, which arises from resource allocation trade-offs at the molecular level and is ultimately reflected in growth and physiology. Elevated temperature primarily upregulated pathways related to carbon and energy metabolism, including glycolysis, the TCA cycle, and lipid biosynthesis, suggesting enhanced energy production and increased carbon storage. In contrast, a high N:P ratio preferentially enhanced the expression of genes involved in nitrogen assimilation, amino acid synthesis, and urea cycle related pathways, indicating allocation toward nitrogen-rich compounds. However, under combined effect of elevated temperature and a high N:P ratio, key urea cycle genes (e.g., *CPS* and *ASS*) were downregulated, whereas GS/GOGAT and carbon metabolism pathways remained upregulated, accompanied by increased light harvesting and constrained electron transport. These molecular reallocations provide a mechanistic basis for the antagonistic physiological responses observed under combined stress, including reduced growth enhancement and lower contents of PON and Chl *a* relative to additive expectations. Previous studies on dinoflagellate blooms have highlighted the importance of GS/GOGAT and carbon metabolism in bloom formation and maintenance, potentially conferring competitive advantage (Zhang et al., 2019b, 2021). From this perspective, the response strategies exhibited by *K. mikimotoi* may enhance its competitive success under the increasingly complex marine environmental changes projected for the future.

5. Conclusions and future directions

This study demonstrates that elevated temperature and a high N:P ratio, both individually and together, significantly influence the physiological and molecular responses of *K. mikimotoi*. Elevated temperature promoted cell growth but reduced Chl *a* and PON contents. The high N:P ratio facilitated nitrogen assimilation, which was evidenced by the stable PON content, increased carbon fixation enzyme activity, and higher ATP content. Under elevated temperature, transcriptomic analyses suggest enhanced energy metabolism and lipid biosynthesis to sustain higher growth rates. Under high N:P conditions, gene expression patterns indicate increased allocation toward the synthesis of nitrogen-rich compounds, including proteins and urea cycle-related pathways, as a strategy for nitrogen storage to support growth and metabolism. In contrast, the combined stress of elevated temperature and a high N:P ratio induced a transcriptionally inferred shift in energy allocation towards soluble sugars and lipids, accompanied by downregulation of urea cycle-related genes, suggesting a trade-off between energy conservation and nitrogen utilization. Interaction analyses also indicated antagonistic effects on growth rate, Chl *a*, and PON content, reflecting complex physiological and transcriptomic trade-offs. Collectively, this study provides novel insights into dinoflagellate responses to environmental changes and underscores the importance of considering the effects of interactive stressors in future global change research. However, this study focused on the short-term response of a single dinoflagellate species, whereas dinoflagellates respond to future environmental changes through long-term adaptation. Future research should investigate the long-term adaptive responses of different dinoflagellate species to understand the evolution and community succession of marine dinoflagellates under global changes.

CRediT authorship contribution statement

Xiang Ye: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Wen-Jing Sun:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Wei-Ping Zhang:** Investigation. **Yang Zhou:** Investigation. **Shuo-Yu Zhang:** Investigation. **Can-Jun Cai:** Investigation. **Shu-Feng Zhang:** Conceptualization. **Jae-Seong Lee:** Conceptualization. **Minghua Wang:** Writing – review & editing, Validation, Supervision. **Da-Zhi Wang:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was partially supported by the National Natural Science Foundation of China (No. 42030404) and the National Key Research and Development Program of China (No. 2022YFC3105302).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.hal.2026.103068.

Data availability

Data will be made available on request.

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